
PROBE-GUIDED GRADIENT BALANCING FOR MULTI-MODAL LEARNING

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ABSTRACT

Jointly trained multimodal networks frequently underutilize their weaker modalities, as faster-learning modalities dominate the shared fusion objective and suppress the gradient signal reaching slower ones. Most existing approaches throttle the dominant modality, whereas directly boosting the weaker modality often fails because monitoring signals derived from the joint loss are entangled with, and thus dominated by, the stronger modality. We propose a new gradient modulation approach, probe-guided gradient boosting (PGGB), to address the problem by probing and boosting the weak modalities while preserving the strength of the dominant modality. Probing is performed by attaching a lightweight linear classifier to each modality’s stop-gradient features to estimate the corresponding representation quality score. The unbiased gap between these scores across modalities, the *utilization gap*, serves as an imbalance signal that defines a bounded and smoothed scaling factor adaptively reweighing the gradients of weaker modalities, enabling stable and effective rebalancing during joint optimization. The method is composable with throttling methods as the intervention leaves the loss and fusion architecture unchanged. Extensive experiments were conducted across eight benchmarks spanning four domains and 2–4 modalities. The results show that our method outperforms state-of-the-art approaches on highly imbalanced multimodal datasets, while remaining competitive on benchmarks with low or no modality imbalance. Bounded scaling, self-attenuation, and a standard-SGD descent bound were established under standard smoothness and bounded-variance assumptions. Code is provided in the supplementary material.

1 INTRODUCTION

Multimodal learning integrates complementary signals across heterogeneous sources, including audio, visual, and textual streams among others (?). Under mild assumptions on latent structure, multimodal models are provably superior to their unimodal counterparts (?). Despite this theoretical promise, jointly trained multimodal networks frequently fail to fully exploit all available modalities, often performing no better than their strongest unimodal component. This *modality laziness* effect (???) has been observed for instance in audio-dominant emotion recognition on the CREMA-D (?) dataset.

Existing methods share a *throttle-only bias*, intervening primarily by damping the dominant modality’s gradient (?????). Some include nominally bidirectional terms (????), but throttling empirically dominates, and ?’s symmetric boost-direction variant fails to close the gap. In late-fusion, every encoder’s gradient flows through a shared classifier whose loss-derivative is dominated by the strong modality’s contribution to the fusion loss. Amplifying the weak encoder’s gradient therefore rescales an already-corrupted shared signal. Recent methods take partial steps (??) but continue to monitor imbalance through signals coupled to the joint training objective.

We propose *probe-guided gradient boosting* (PGGB): lightweight linear probes (?) on stop-gradient features (the stop-gradient operator passes activations forward but blocks the gradient in the backward pass (?)) yield a per-modality representation-quality estimate, whose cross-modality spread defines the *utilization gap*, independent of the shared classifier’s loss-derivative, removing the chain-rule coupling that impaired prior boost-the-weak attempts (?). The signal drives a bounded multiplicative factor on the weaker encoder’s gradient, smoothed by an exponential moving average (EMA) (?). Because the factor acts only on encoder gradients and modifies neither the loss nor the fusion

architecture, it is *composable* with throttling methods, multiplying with their suppression factors to produce two-sided pressure toward balance.

We evaluate PGGB on eight benchmarks spanning audio-visual, sentiment, text-image, and medical-segmentation tasks (details in Section ??). The main gain is observed on the high-imbalance CREMA-D benchmark (3-frame), where composing PGGB with a representative gradient-modulation baseline (?) reaches $71.45 \pm 1.71\%$ accuracy, +2.31 pp over the same baseline alone and +9.86 pp over joint training. On the four low-imbalance benchmarks (AVE, Kinetics-Sounds, Twitter15, Sarcasm), PGGB alone is the best-performing method on all four (Table ??). On the high-imbalance text-dominant sentiment benchmarks (CMU-MOSI, CMU-MOSEI), where text alone is near-multimodally sufficient and prior balancing methods report only modest gains (??), PGGB+OGM-GE remains within seed standard deviation of the strongest baseline (−0.08 and −0.04 pp). The OGM-GE composition variant has two isolated regressions versus joint training (Kinetics-Sounds −1.72 pp, Sarcasm −0.64 pp) noted in Section ??, while several gradient-modulation baselines degrade on multiple datasets (e.g., this baseline (?) regresses on Kinetics-Sounds by −1.80 pp, Table ??).

Contributions.

- We identify a *throttle-only bias* in the modality-balancing literature and show that its natural complement, boosting the weaker modality, fails when the monitoring signal is coupled to the joint training objective via shared loss derivatives in late fusion.
- We propose *decoupled probe monitoring*, a diagnostic attaching lightweight linear probes to stop-gradient features, yielding a unbiased *utilization gap* estimate (the spread of per-modality probe accuracies) independent of the joint objective. The decoupling enables effective weak-modality boosting where prior coupled-monitoring boost-the-weak attempts failed (?).
- We introduce *PGGB*, one instantiation that converts the monitoring signal into a bounded multiplicative scaling on the weaker encoder’s gradients. It is composable with existing throttling methods (Appendix ?? validates non-destructive composition with four such methods).
- We validate *PGGB* on eight benchmarks spanning audio-visual, sentiment, text-image, and medical-segmentation tasks. *PGGB* composed with throttling outperforms state-of-the-art baselines on high-imbalance audio-visual (CREMA-D), matches the strongest available baseline on medical segmentation (BraTS 2021, +0.28 pp, within seed variance), and remains competitive within seed variance on the high-imbalance text-dominant sentiment benchmarks (CMU-MOSI, CMU-MOSEI), and *PGGB* used alone is the best-performing method on all four low-imbalance datasets (AVE, Kinetics-Sounds, Twitter15, Sarcasm).

2 RELATED WORK

Modality imbalance in joint training. Jointly trained multimodal networks frequently underperform their best unimodal component (?). Under a Gaussian mixture model, ? prove that joint training learns only a subset of modalities and derive bounds on the duration of the resulting unimodal phase, and ? characterize the conditions under which individual modality features fail to be learned. ? further show, in deep linear networks, that this unimodal phase depends on architecture depth and initialization. Modality imbalance is therefore a structural property of joint optimization, rooted in the shared loss through which all encoder gradients flow.

Gradient modulation for modality balance. The dominant response has been to modulate gradient magnitudes during backpropagation. Multi-task precursors GradNorm (?) and PCGrad (?) inspired a line of multimodal work. OGM-GE (?) dampens the dominant modality’s gradient via a ratio of per-modality contributions to the target-class score, refined by AGM (?), CGGM (?), and MLGM (?). Although some methods include bidirectional terms, throttling empirically dominates and boost-only variants remain ineffective: the symmetric boost direction (OGM*) reported in ? fails to close the modality gap, which we attribute to coupled monitoring signals affected by the dominant modality (Section ??). M-SAM (?) pairs Shapley attribution with sharpness-aware perturbation, inheriting the same coupled-monitoring limitation.

Alternative approaches to modality balance. Beyond gradient scaling, a broader set of interventions has been proposed. These include feed-forward feature modulation (?) and Pareto-optimal aggregation (?). Modality-alternating schemes interleave unimodal and joint updates (?). Diagnostic methods detect imbalance through separability (?), Fisher information (?), or bias-variance decomposition (?). Other approaches intervene at the data level (?). Per-modality learning-rate methods (?) are an adjacent class operating on the optimizer step rather than the gradient. Despite their diversity, these methods share a common design choice. Imbalance is detected through signals *coupled* to the joint objective, such as gradient ratios, per-modality losses, or live-model predictions (with the partial exception of PMR’s decoupled unimodal prototypes (?)). These are precisely the quantities distorted by dominant-modality dynamics, which limits the reliability of the resulting corrections.

Representation monitoring and probing. Linear probes on frozen representations (?) are a standard tool for assessing representation quality. Several multimodal methods adopt probe-like monitoring but retain coupling to the joint objective. IPRM’s two-pass forward strategy compares unimodal and fused predictions of the live model via KL divergence (?), CGGM’s auxiliary classifiers participate in the training loss (?), and AUG (?) relies on loss-based weakness signals that inherit the dominant modality’s bias. PMR (?) moves toward decoupling via unimodal prototypes but the prototype signal feeds a gradient coefficient on the shared encoder, while DI-MML (?) detaches only the unimodal pre-fusion stage. In contrast, we train *fully decoupled* linear probes on stop-gradient features with separate optimizers and evaluate them on held-out within-batch samples to prevent probe overfitting. The resulting utilization gap directly boosts the weaker encoder’s gradients, so the imbalance estimate is not affected by the dynamics it aims to correct.

3 PROBE-GUIDED GRADIENT BOOSTING

We present PGGB in four stages: problem formulation (§??), decoupled probe monitoring (§??), the boost mechanism (§??), and composability with existing balancing methods (§??). Figure ?? traces the data flow: the top panel depicts the forward path (§??), the bottom panel depicts probe monitoring on stop-gradient features (§??), and the green boost block plus actuation rail depict the gradient-modification mechanism (§??). Composition with throttling is described in (§??).

3.1 PROBLEM FORMULATION

Each multimodal sample comprises M inputs $\{\mathbf{x}_m\}_{m=1}^M$, one per modality. Encoder f_m with parameters θ_m maps $\mathbf{x}_m$ to $\mathbf{z}_m$:

$$\mathbf{z}_m = f_m(\mathbf{x}_m; \theta_m) \in \mathbb{R}^d. \quad (1)$$

Each $\mathbf{z}_m$ is a real value feature vector in $\mathbb{R}^d$, with d common across modalities.

$$\mathcal{L}_{\text{fusion}} = \ell(\hat{y}, y). \quad (2)$$

Here ℓ is a differentiable per-sample loss mapping a prediction-target pair to a non-negative scalar. By the chain rule, the gradient with respect to encoder parameters decomposes as

$$\partial \mathcal{L}_{\text{fusion}} / \partial \theta_m = (\partial \mathcal{L}_{\text{fusion}} / \partial g) (\partial g / \partial \mathbf{z}_m) (\partial \mathbf{z}_m / \partial \theta_m). \quad (3)$$

The term $\partial \mathcal{L}_{\text{fusion}} / \partial g$ is shared across all modalities, since all encoder outputs flow through the same fusion head. This shared dependency is the root of coupled monitoring failure: when one modality (say $m=1$) converges faster and begins to dominate the head’s outputs, $\partial \mathcal{L}_{\text{fusion}} / \partial g$ increasingly reflects the dominant modality’s contribution. Any monitoring signal derived from $\mathcal{L}_{\text{fusion}}$ or its gradients, including the discrepancy ratio used by OGM-GE (?) and the auxiliary classifier losses used by CGGM (?), inherits this bias.

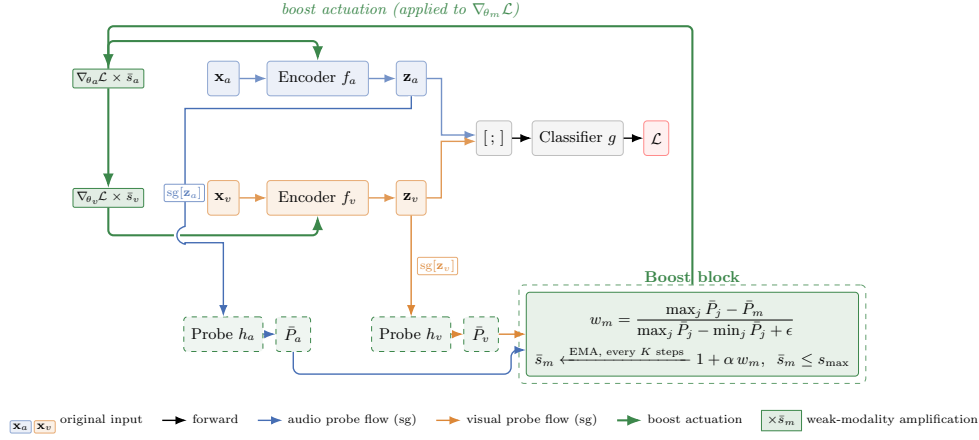


Figure 1: An Illustration of probe-guided gradient boosting on 2 modalities. **Top panel:** features $\mathbf{z}_m$ are concatenated and classified via the shared fusion classifier g , producing $\mathcal{L}$. **Bottom panel:** stop-gradient features $\text{sg}[\mathbf{z}_m]$ are evaluated by linear probes along solid per-modality probe-flow lines (audio in blue, visual in orange), yielding per-modality accuracies $\bar{P}_m$. The boost block computes the relative-gap weight w_m and a smoothed scale $\bar{s}_m$ (EMA-smoothed, refreshed only at probe-evaluation events every K steps and held constant in between), clipped at $s_{\max}$. The actuation rail (thick green) applies $\bar{s}_m$ to the per-modality gradient $\nabla_{\theta_m} \mathcal{L}$. Probes never backpropagate into encoders, so the monitoring signal is structurally decoupled from the fusion objective.

3.2 DECOUPLED PROBE MONITORING

We attach a lightweight linear probe classifier $h_m: \mathbb{R}^d \rightarrow \mathbb{R}^C$ (with C outputs) to each encoder, parameterized by ψ_m . The resulting probe accuracies define an imbalance signal, the *utilization gap* (Eq. ??), independent of the joint training objective. The probes are trained on stop-gradient features $\text{sg}[\mathbf{z}_m]$, where $\text{sg}[\cdot]$ denotes the stop-gradient operator (?) (identity at the forward pass, zero partial derivatives at the backward pass):

$$\mathcal{L}_{\text{probe},m} = \ell_{\text{probe}}(h_m(\text{sg}[\mathbf{z}_m]; \psi_m), y). \quad (4)$$

Each probe has its own optimizer that updates only ψ_m , and no gradient flows into θ_m . Since $\text{sg}[\mathbf{z}_m]$ halts backward propagation at the feature level, $\mathcal{L}_{\text{probe},m}$ depends only on the quality of $\mathbf{z}_m$ as a representation, not on how it contributes through g . This is precisely why decoupling matters: the probe accuracy P_m measures representation quality not influenced by the dominant modality as the influence can be both ways.

Split-batch evaluation and on-the-fly estimation. Every K training iteration, we partition the (shuffled) mini-batch into two disjoint halves, training probes on the first half and evaluating on the second. This yields a unbiased estimate P_m of the current probe’s training-distribution accuracy (conditioned on $\psi_m^{(t)}$), the evaluation half is held out from probe training and drawn i.i.d. from the training distribution. To convert this per-evaluation sample into a stable estimator, we accumulate an exponential moving average (EMA):

$$\bar{P}_m \leftarrow \beta \cdot P_m + (1 - \beta) \cdot \bar{P}_m, \quad (5)$$

where $\beta \in [0, 1]$ is the smoothing coefficient and other low-pass smoothing methods apply. At $\beta = 0.1$ (§??), the steady-state variance is $\beta/(2 - \beta) \sigma^2 \approx 0.053 \sigma^2$ under an i.i.d. assumption with per-sample variance σ^2 , approximately $19\times$ smaller than a single-batch estimate. The *utilization gap* is the spread of smoothed probe accuracies:

$$\delta = \max_m \bar{P}_m - \min_m \bar{P}_m. \quad (6)$$

Appendix ?? verifies empirically that the EMA estimator substantially reduces mini-batch noise in the probe signal during training.

3.3 PROBE-GUIDED GRADIENT BOOSTING

Given the EMA-smoothed probe accuracies $\{\bar{P}_m\}_{m=1}^M$ (Eq. ??), we compute a per-modality boost scale that amplifies the gradients of weaker encoders (full pseudocode in Appendix ??, Algorithm ??). The procedure involves three steps.

Step 1: Relative weakness. For each modality m , we compute a normalized weakness score $w_m \in [0, 1]$ (zero for the strongest modality, one for the weakest):

$$w_m = \frac{\max_j \bar{P}_j - \bar{P}_m}{\max_j \bar{P}_j - \min_j \bar{P}_j + \epsilon}, \quad (7)$$

where j ranges over all modalities $\{1, \dots, M\}$, and ϵ is a small positive constant serving as a numerical guard against division by zero.

Step 2: Boost scale. w_m is converted to a multiplicative scaling factor:

$$s_m = \min(1 + \alpha \cdot w_m, s_{\max}), \quad (8)$$

where $\alpha > 0$ controls the boost strength and $s_{\max} > 1$ caps the maximum amplification to prevent extreme scaling. When modalities are balanced (δ small), weakness scores become near zero and $s_m \approx 1$, so the method applies no intervention (Proposition ?? gives the precise continuity bound).

Step 3: Smoothing and gradient modification. To avoid abrupt changes in the boost signal, we apply temporal smoothing to the scales. We use a separate exponential moving average (EMA), distinct from the probe-accuracy EMA in Eq. (??), though our analysis (Propositions ?? and ??) is stated specifically for this EMA:

$$\bar{s}_m \leftarrow \mu \cdot s_m + (1 - \mu) \cdot \bar{s}_m, \quad (9)$$

where $\mu \in (0, 1)$ is the EMA smoothing coefficient on the boost scales (set in Section ??). The smoothed scales are applied to the encoder gradients *after* the backward pass and *before* the optimizer step:

$$\frac{\partial \mathcal{L}}{\partial \theta_m} \leftarrow \bar{s}_m \cdot \frac{\partial \mathcal{L}}{\partial \theta_m}. \quad (10)$$

3.4 COMPOSABILITY WITH GRADIENT THROTTLING

Because PGGB amplifies weaker modalities without modifying the dominant modality’s gradient, it is mechanistically compatible with synchronous balancing methods such as gradient throttling, loss reweighting, and Pareto rescaling, with the empirical magnitude of the compound gain depending on the specific base method (Section ??, Appendix ??). The main experiments compose with OGM-GE (?), and Appendix ?? validates composition with four additional methods.

OGM-GE damps the dominant modality’s gradient by a *coupled* signal (inter-modal gradient-magnitude ratio ρ derived from $\mathcal{L}_{\text{fusion}}$), whereas our boost uses a *decoupled* signal (probe accuracy on stop-gradient features $\text{sg}[\mathbf{z}_m]$). The two mechanisms compose multiplicatively on encoder gradients:

$$\frac{\partial \mathcal{L}}{\partial \theta_m} \leftarrow \bar{s}_m \cdot \omega_m^{\text{ogm}} \cdot \frac{\partial \mathcal{L}}{\partial \theta_m}, \quad (11)$$

where $\omega_m^{\text{ogm}} \leq 1$ is the OGM-GE throttle and $\bar{s}_m \geq 1$ is our boost scale.

Following established practice for auxiliary unimodal supervision (?), we optionally include unimodal auxiliary losses, yielding the total objective

$$\mathcal{L} = \mathcal{L}_{\text{fusion}} + \gamma \sum_{m=1}^M \ell_{\text{aux}}(g_m(\mathbf{z}_m), y), \quad (12)$$

where γ is a scalar weight on the per-modality auxiliary terms (set in Section ??). ℓ_{aux} matches the form of ℓ (Eq. ??) for each benchmark.

3.5 SAFETY PROPERTIES

We establish three safety properties: bounded scaling, self-attenuation, and a standard-SGD descent bound. These are structural guarantees that the method does not diverge or harm low-imbalance training, not mechanism-specific convergence results. A rate-level analysis tying gap closure to probe dynamics is left for future work. Full proofs are in Appendix ??.

Assumption 1 (Smoothness). *The loss $\mathcal{L}(\theta)$ is L -smooth: $\|\nabla \mathcal{L}(\theta) - \nabla \mathcal{L}(\theta')\| \leq L\|\theta - \theta'\|$ for all θ, θ' .*

Assumption 2 (Bounded variance). *The stochastic gradient $\mathbf{g}(\theta)$ satisfies $\mathbb{E}[\mathbf{g}(\theta)] = \nabla \mathcal{L}(\theta)$ and $\mathbb{E}[\|\mathbf{g}(\theta) - \nabla \mathcal{L}(\theta)\|^2] \leq \sigma^2$ for all θ .*

Proposition 1 (Bounded scaling). *For all modalities m and all iterations t , the smoothed boost scale satisfies $\bar{s}_m^{(t)} \in [1, s_{\max}]$.*

Proposition 2 (Self-attenuation). *For any $\Delta \geq 0$, if the smoothed probe accuracies are Δ -balanced ($\max_j \bar{P}_j - \min_j \bar{P}_j \leq \Delta$) at iteration t , then each modality’s instantaneous boost scale satisfies $|\bar{s}_m^{(t)} - 1| \leq \alpha\Delta/(\Delta + \epsilon)$, and the EMA-smoothed scale converges geometrically to unity at rate $(1 - \mu)$ as long as this condition persists.*

Proposition 3 (Descent bound). *Under Assumptions ?? and ??, SGD with PGGB and constant learning rate $\eta \leq 1/(L \cdot s_{\max}^2)$ satisfies*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2] \leq \frac{2\Delta_{\mathcal{L}}}{\eta T} + L\eta s_{\max}^2 \sigma^2, \quad (13)$$

where $\Delta_{\mathcal{L}} = \mathcal{L}(\theta_0) - \mathcal{L}^*$ is the initial optimality gap and $\mathcal{L}^*$ is a lower bound on $\mathcal{L}$.

The bound matches the standard nonconvex SGD rate (?) up to an $s_{\max}^2$ factor in the variance term, a controlled increase ($s_{\max} = 2$), and Proposition ?? ensures this cost is incurred only when imbalance is present. The smoothness constant L in Assumption ?? is loss-dependent, determined by the choice of ℓ in Eq. (??); in our experiments, the per-task learning rate (Section ??) is selected consistent with the descent inequality for the corresponding ℓ .

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets. We evaluate PGGB on eight benchmarks spanning three tasks and 2–4 modalities, reported as mean $\pm$ std over five seeds. *Audio-visual:* CREMA-D (?) (3-frame at 3 fps, following OGM-GE), AVE (?), Kinetics-Sounds (KS) (?). *Sentiment:* CMU-MOSEI (?), CMU-MOSI (?). *Text-image:* Twitter15 (??), Sarcasm (?). *Segmentation:* BraTS 2021 (?). Encoders follow task grouping (ResNet-18 for audio-visual, MLPs on pre-extracted features for sentiment and text-image, DeepLab v3+/ResNet-101 for BraTS), with splits and caveats in Appendix ?. We assume each modality is task-relevant, consistent with these benchmarks’ established multimodal use; PGGB does not validate relevance and would amplify a noise-only modality if one were present.

Baselines. We compare experiment results against eight methods: **Baseline** (joint training, no balancing), **OGM-GE** (?) (discrepancy-ratio gradient throttling), **MMPareto** (?) (Pareto-optimal gradient weights), **AGM** (?) (Shapley-based adaptive gradient modulation), **G-Blend** (?) (OG-ratio loss weighting), **CGGM** (?) (classifier-guided magnitude/direction modulation¹), **InfoReg** (?) (unimodal information regularization), and **MILES** (?) (per-modality learning-rate adjustment). InfoReg and MILES are evaluated on CREMA-D only, as their official implementations target audio-visual classification. MMPareto, AGM, and G-Blend are evaluated on all classification benchmarks but not on BraTS 2021, as their official implementations target classification losses and do not support dense-prediction multi-class Dice objectives out of the box.

Reproduction protocol. All baselines are re-run under a matched pipeline (shared encoders, 3-frame CREMA-D, identical augmentation and optimizer) for fair comparison. OGM-GE (69.14%) is

¹Adapted from its original Transformer setting to our CNN/MLP pipeline. The reproduction shows consistent regression across all eight benchmarks, including off-table Food101 (−37 pp vs. baseline, Appendix ??), which we attribute to backbone mismatch rather than the underlying method.

3–6 pp below the 72–75% in follow-up work using different frame sampling, AGM (57.42%) uses our ResNet-18 rather than its original backbone, MILES is ported to our matched ResNet-18 pipeline from its specialized backbone, and CGGM from its Transformer setting. Sub-baseline AGM/MILES numbers plausibly reflect 3-frame protocol sensitivity, and published configurations are retained without retuning.

Implementation details. Classification uses concatenation fusion, batch size 64, 100 epochs. Audio-visual datasets use SGD (learning rate 0.001, weight decay 10^{-4}) with StepLR, sentiment and text-image use Adam (learning rate 0.001) with StepLR (step 40), and BraTS uses SGD with cosine scheduling (base learning rate 0.01, 5-epoch warmup) and batch size 12. PGGB uses identical hyperparameters across all datasets: $K=20$, $\beta=0.1$, $\mu=0.3$, $s_{\max}=2.0$, $\gamma=1.0$, and $\alpha \in \{0.5, 0.75\}$ for the standalone and OGM-GE-combined variants respectively. Probes are trained with Adam (learning rate 10^{-3}).

Computational overhead. Each linear probe $h_m: \mathbb{R}^d \rightarrow \mathbb{R}^C$ adds at most a few thousand parameters per modality, negligible relative to the encoders (e.g., ResNet-18 at $\sim 11\text{M}$ parameters). Probe training every $K=20$ iterations on a half mini-batch and the multiplicative gradient scaling at the optimizer step together add approximately 1% wall-clock overhead relative to the baseline. The EMA state $\{\bar{P}_m, \bar{s}_m\}$ requires $O(M)$ additional scalars.

Imbalance categorization. We categorize benchmarks by the utilization gap δ (Eq. (??)) measured during baseline training, with per-dataset values in Appendix ??: *high-imbalance* ($\delta > 0.15$, CREMA-D, CMU-MOSI, CMU-MOSEI) and *low-imbalance* ($\delta \leq 0.15$, AVE, KS, Twitter15, Sarcasm). BraTS 2021 is reported separately as a dense-prediction segmentation benchmark. The high-imbalance group is heterogeneous in source: CREMA-D imbalance arises from audio-dominance in emotion recognition (?), whereas CMU-MOSI/MOSEI imbalance reflects text-sufficiency, with text-only baselines within roughly 1 pp of multimodal accuracy on CMU-MOSEI (?) and only a +2.7 pp multimodal gain on CMU-MOSI (?). On these text-dominant benchmarks, balancing methods report only modest absolute gains (?), motivating within-seed-std competitiveness as the relevant metric.

Table 1: Main results across eight benchmarks. We report accuracy (%) for classification datasets and average Dice (%) for BraTS 2021. All values are mean $\pm$ std over 5 seeds. **Bold** indicates the best result per dataset. Imbalance categorization is described in Section ??.

Method	CREMA-D	AVE	KS	CMU-MOSI	Twitter15	Sarcasm	CMU-MOSEI	BraTS 2021
Baseline	61.59 $\pm$ 0.80	86.54 $\pm$ 0.42	79.05 $\pm$ 0.40	72.42 $\pm$ 0.48	66.60 $\pm$ 0.57	82.40 $\pm$ 0.17	70.42 $\pm$ 0.29	85.77 $\pm$ 0.62
OGM-GE	69.14 $\pm$ 1.13	86.96 $\pm$ 0.71	77.25 $\pm$ 0.79	72.68 $\pm$ 0.89	66.33 $\pm$ 0.21	81.81 $\pm$ 0.23	72.47 $\pm$ 0.70	86.21 $\pm$ 0.17
MMPareto	65.51 $\pm$ 0.87	86.37 $\pm$ 0.33	78.21 $\pm$ 0.38	72.68 $\pm$ 0.79	66.81 $\pm$ 0.47	82.27 $\pm$ 0.12	70.20 $\pm$ 0.71	—
AGM	57.42 $\pm$ 0.73	84.42 $\pm$ 0.26	77.85 $\pm$ 0.30	72.50 $\pm$ 0.62	66.79 $\pm$ 0.47	82.32 $\pm$ 0.16	69.28 $\pm$ 0.67	—
G-Blend	61.10 $\pm$ 1.87	87.09 $\pm$ 0.29	77.75 $\pm$ 1.13	72.36 $\pm$ 0.66	66.66 $\pm$ 0.57	82.35 $\pm$ 0.26	70.15 $\pm$ 0.66	—
CGGM	50.22 $\pm$ 1.39	76.72 $\pm$ 0.42	73.18 $\pm$ 0.27	59.45 $\pm$ 0.36	62.41 $\pm$ 0.36	80.71 $\pm$ 0.27	68.05 $\pm$ 0.46	81.13 $\pm$ 1.34
InfoReg	67.72 $\pm$ 0.83	—	—	—	—	—	—	—
PGGB ($\alpha=0.5$)	62.72 $\pm$ 1.65	87.41 $\pm$ 0.26	79.17 $\pm$ 0.97	71.89 $\pm$ 0.82	66.85 $\pm$ 0.50	82.44 $\pm$ 0.35	69.80 $\pm$ 0.80	85.98 $\pm$ 1.15
PGGB+OGM-GE ($\alpha=0.75$)	71.45 $\pm$ 1.71	87.23 $\pm$ 0.58	77.33 $\pm$ 0.64	72.60 $\pm$ 0.94	66.59 $\pm$ 0.51	81.76 $\pm$ 0.08	72.43 $\pm$ 0.65	86.49 $\pm$ 0.21

4.2 MAIN RESULTS

Table ?? reports results across eight benchmarks. On high-imbalance CREMA-D, PGGB+OGM-GE attains 71.45%, +2.31 pp over OGM-GE and +9.86 pp over baseline, the best among competitors. A three-way decomposition in §?? isolates this gain to the boost actuation, with the decoupled monitor causally inert at $\alpha=0$. PGGB alone (+1.13 pp) is insufficient under severe imbalance, confirming that two-sided intervention is required. On the four low-imbalance benchmarks (AVE, KS, Twitter15, Sarcasm, with cross-class MSLR comparison in Appendix ??), PGGB alone is the best-performing method on all four, whereas OGM-GE often regresses (KS -1.80 pp, Twitter15 -0.27 pp, Sarcasm -0.59 pp), consistent with self-attenuation as $\delta \rightarrow 0$ (§??). On the high-imbalance text-dominant sentiment benchmarks (CMU-MOSI, CMU-MOSEI), where text alone is near-multimodally sufficient and prior balancing methods report only modest absolute gains (?), PGGB+OGM-GE remains within seed standard deviation of the strongest baseline (MOSI -0.08 pp, MOSEI -0.04 pp). On BraTS 2021 (DeepLab v3+/ResNet-101, Dice loss), PGGB+OGM-GE yields 86.49% Dice (+0.72 pp), extending the mechanism to dense prediction. The composability claim further extends on AVE to two of four non-OGM-GE methods tested (Appendix ??). A comparison against OPM (?), whose single-layer-fusion dependency requires a separate protocol, is in Appendix ??.

4.3 ABLATION STUDIES

We ablate on CREMA-D with single-frame visual input, which widens the modality gap and amplifies method differences (Table ??).

Table 2: Ablation on CREMA-D (1-frame). Accuracy (%), mean $\pm$ std over 5 seeds. The upper row reports the baseline. Lower rows isolate single-mechanism interventions (boost only, OGM-GE only) and their combinations.

Method	Accuracy (%)
Baseline	60.30 $\pm$ 0.70
PGGB ($\alpha=0.5$)	60.46 $\pm$ 0.85
OGM-GE only	62.47 $\pm$ 1.42
PGGB+OGM-GE ($\alpha=0.5$)	62.37 $\pm$ 0.57
PGGB+OGM-GE ($\alpha=0.75$)	62.69 $\pm$ 0.22

Among single-mechanism interventions, OGM-GE alone yields +2.17 pp versus PGGB alone +0.16 pp, consistent with prior findings that throttling is the more impactful individual intervention (?). Combining PGGB with OGM-GE at $\alpha=0.75$ achieves the best result (62.69%) with a sixfold reduction in std (0.22% vs. 1.42%), indicating that decoupled probe monitoring provides a stabilizing signal complementing the noisier gradient-ratio throttling.

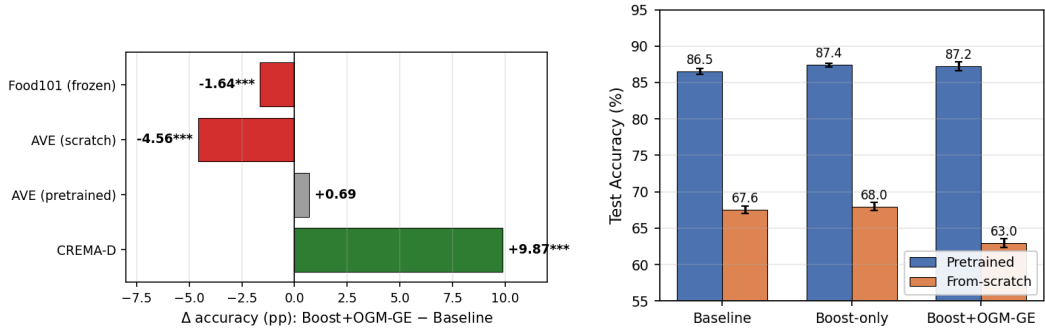
Isolating the monitor from the boost. We run three 5-seed decompositions on CREMA-D 3-frame (Table ?? protocol): (i) $\alpha=0$ with probes active (69.14 $\pm$ 1.13%, reported as OGM-GE alone in Table ??), (ii) an independent 5-seed replication of the same $\alpha=0$ condition (69.14 $\pm$ 1.18%), and (iii) $\alpha=0.75$ (71.45 $\pm$ 1.71%). Conditions (i) and (ii) are operationally identical (identical α , identical probe pipeline) and yield the same mean within 0.00 pp, confirming estimate stability. The +2.31 pp lift at (iii) isolates boost actuation: it cannot be attributed to probe-pipeline side-effects matched across conditions (probe training on stop-gradient features, optimizer steps on probe parameters, and EMA state updates). Only the multiplicative scale s_m differs between $\alpha=0$ and $\alpha=0.75$.

4.3.1 OPERATING CONDITIONS ABLATION

A complementary question to *which* component drives the CREMA-D gain is *when* the method is effective. We characterize four regimes along three axes (feature learning, encoder initialization, data-to-architecture ratio) in Table ?? and Figure ??.

Table 3: Operating conditions across four regimes. Columns: trainable encoders, modality asymmetry (pretraining or LR differences), adequate data-to-architecture ratio, and complementary modalities. Δ : accuracy change (pp), PGGB+OGM-GE vs. baseline. Significance: *** $p < 0.001$ (Welch’s t-test, 5 seeds per condition). AVE (from-scratch) is the clearest failure case (−4.56 pp), with the pretrained $\leftrightarrow$ scratch contrast localizing sensitivity to the OGM-GE component. Per-regime analysis in Appendix ??.

Regime	Trainable	Asymmetry	Adequate data	Compl.	Δ
CREMA-D	✓	✓	✓	✓	+9.86***
AVE (pretrained)	✓	✓ (pretrain)	✓	✓	+0.69
AVE (from-scratch)	✓	×	× ($\sim$ 4K)	✓	−4.56***
Food101 (frozen)	×	—	✓	✓	−1.64



(a) Δ accuracy (PGGB+OGM-GE vs. baseline) across four regimes. (b) AVE test accuracy: pretrained vs. from-scratch encoders.

Figure 2: Operating-conditions ablation (5 seeds). (a) PGGB alone self-attenuates toward baseline when Table ?? conditions are violated, whereas PGGB+OGM-GE inherits OGM-GE’s regime-dependent failure modes (AVE from-scratch is the clearest case). (b) With data and architecture fixed, flipping initialization from pretrained to random shifts PGGB+OGM-GE from +0.69 to −4.56 pp ($p=0.0001$) while PGGB alone stays positive, localizing sensitivity to the OGM-GE component.

Encoder initialization and scope. Fixing dataset and architecture but varying encoder initialization gives the sharpest contrast (Figure ??). PGGB alone stays positive in both regimes (+0.87/+0.42 pp, both $p < 0.05$), while PGGB+OGM-GE flips sign (+0.69 $\rightarrow$ −4.56 pp, $p=0.0001$). ImageNet pretraining installs an asymmetry aligned with OGM-GE’s throttling assumption. Under from-scratch initialization both encoders are simultaneously under-trained on $\sim 4K$ samples, so suppressing either destabilizes the joint dynamic. This localizes the failure to the OGM-GE component (Appendix ??).

4.4 ANALYSIS

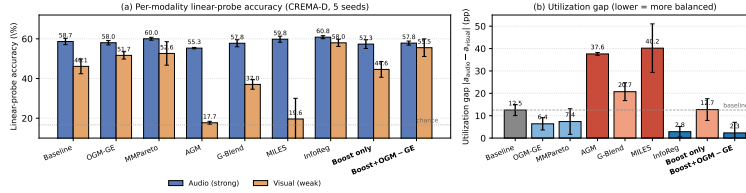


Figure 3: Per-modality diagnostics on CREMA-D (5 seeds). (a) Post-hoc linear-probe accuracy P_m^{post} (Appendix ??). Dotted line marks chance (1/6). AGM and MILES collapse the weak probe. (b) Utilization gap $|P_a^{\text{post}} - P_v^{\text{post}}|$. PGGB+OGM-GE attains 2.31 pp ($5.4\times$ reduction vs. baseline gap). Error bars: ± 1 std.

We probe per-modality utilization with linear classifiers trained on frozen encoder features. Post-hoc linear probes (Appendix ??, Figure ??b) show PGGB+OGM-GE attains the smallest utilization gap (2.31 pp): the weak probe rises +9.41 pp, the strong falls only −0.81 pp, while AGM, MILES, and G-Blend degrade the weak probe. Extension to M modalities uses pairwise weakness (Eq. ??).

5 CONCLUSION

We propose PGGB, a method for balanced multimodal learning that amplifies the contribution of weaker modalities while preserving stronger ones, composing naturally with existing throttling techniques. PGGB alone is the best-performing method on all four low-imbalance benchmarks. On high-imbalance CREMA-D, composition with OGM-GE yields +2.31 pp over the strongest baseline, and on the text-dominant sentiment benchmarks (CMU-MOSI, CMU-MOSEI) the same composition remains within seed standard deviation of the strongest baseline. On medical segmentation (BraTS 2021), it matches the strongest available baseline (+0.28 pp, within seed variance). **Limitations.** Per-setting tuning is required, probes add small compute overhead, and composition with throttling can regress when the dominance gap is small (worst case AVE-from-scratch −4.56 pp, §??), a regime our theory does not cover. **Future work.** Probe-driven adaptive composition that engages throttling only when the dominance gap warrants, and extension beyond classification and segmentation. No new data is introduced, and §?? should precede deployment.

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A PROOFS OF CONVERGENCE PROPERTIES

We restate and prove each proposition from Section ?? . All notation follows the main text: $\mathcal{L}$ denotes the training loss, $\mathbf{g}(\theta)$ a stochastic gradient, $\bar{s}_m$ the EMA-smoothed boost scale for modality m , and $s_{\max}$ the scale cap.

Proposition 4 (Bounded scaling, restated). *For all modalities m and all iterations $t \geq 0$, the smoothed boost scale satisfies $\bar{s}_m^{(t)} \in [1, s_{\max}]$.*

Proof. We proceed by induction on t . At initialization, $\bar{s}_m^{(0)} = 1 \in [1, s_{\max}]$, establishing the base case.

For the inductive step, suppose $\bar{s}_m^{(t)} \in [1, s_{\max}]$. By Eq. (??), the weakness score satisfies $w_m \in [0, 1]$ since the numerator $\max_j \bar{P}_j - \bar{P}_m$ lies in $[0, \max_j \bar{P}_j - \min_j \bar{P}_j]$ and the denominator equals $\max_j \bar{P}_j - \min_j \bar{P}_j + \epsilon$. Substituting into Eq. (??):

$$s_m = \min(1 + \alpha \cdot w_m, s_{\max}) \in [1, s_{\max}],$$

since $w_m \geq 0$ implies $1 + \alpha \cdot w_m \geq 1$, and the min operation enforces the upper bound.

The EMA update (Eq. (??)) gives:

$$\bar{s}_m^{(t+1)} = \mu \cdot s_m + (1 - \mu) \cdot \bar{s}_m^{(t)}.$$

Since $\mu \in (0, 1)$, this is a convex combination of $s_m \in [1, s_{\max}]$ and $\bar{s}_m^{(t)} \in [1, s_{\max}]$ (by the inductive hypothesis), so $\bar{s}_m^{(t+1)} \in [1, s_{\max}]$. $\square$

Proposition 5 (Self-attenuation, restated). *For any $\Delta \geq 0$, if $\max_j \bar{P}_j^{(t)} - \min_j \bar{P}_j^{(t)} \leq \Delta$ at all iterations $t \geq t_0$, then*

1. $|s_m^{(t)} - 1| \leq \alpha \cdot \Delta / (\Delta + \epsilon)$ for all m and all $t \geq t_0$, and
2. $|\bar{s}_m^{(t)} - 1| \leq (1 - \mu)^{t-t_0} \cdot (s_{\max} - 1) + \alpha \cdot \Delta / (\Delta + \epsilon)$.

In particular, when $\Delta \rightarrow 0$ we recover $s_m = 1$ exactly, and the EMA-smoothed scale converges to unity at rate $(1 - \mu)$.

Proof. Let $\delta^{(t)} := \max_j \bar{P}_j^{(t)} - \min_j \bar{P}_j^{(t)}$. By assumption $\delta^{(t)} \leq \Delta$ for all $t \geq t_0$. From Eq. (??), the weakness score satisfies

$$w_m = \frac{\max_j \bar{P}_j - \bar{P}_m}{\max_j \bar{P}_j - \min_j \bar{P}_j + \epsilon} = \frac{\max_j \bar{P}_j - \bar{P}_m}{\delta + \epsilon} \leq \frac{\Delta}{\Delta + \epsilon},$$

since the numerator is bounded by $\delta \leq \Delta$ and the denominator is non-decreasing in δ . Then by Eq. (??),

$$s_m = \min(1 + \alpha w_m, s_{\max}) \in [1, 1 + \alpha \cdot \frac{\Delta}{\Delta + \epsilon}],$$

which gives part (1) immediately. For part (2), the EMA update (Eq. (??)) is

$$\bar{s}_m^{(t+1)} - 1 = \mu(s_m^{(t)} - 1) + (1 - \mu)(\bar{s}_m^{(t)} - 1),$$

so by the triangle inequality and induction from t_0 ,

$$|\bar{s}_m^{(t)} - 1| \leq (1 - \mu)^{t-t_0} |\bar{s}_m^{(t_0)} - 1| + \mu \sum_{k=0}^{t-t_0-1} (1 - \mu)^k \cdot \max_{t_0 \leq \tau < t} |s_m^{(\tau)} - 1|.$$

Using $|\bar{s}_m^{(t_0)} - 1| \leq s_{\max} - 1$ (Proposition ??), $|s_m^{(\tau)} - 1| \leq \alpha \Delta / (\Delta + \epsilon)$, and the partial sum $\mu \sum_{k=0}^{t-t_0-1} (1 - \mu)^k = 1 - (1 - \mu)^{t-t_0} \leq 1$, the bound follows. In the limit $\Delta \rightarrow 0$ the second term vanishes and the deviation decays geometrically at rate $(1 - \mu)$. $\square$

Proposition 6 (Convergence, restated). *Under Assumptions ?? and ??, SGD with PGGB and constant learning rate $\eta \leq 1/(L \cdot s_{\max}^2)$ satisfies*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2] \leq \frac{2\Delta_{\mathcal{L}}}{\eta T} + L\eta s_{\max}^2 \sigma^2,$$

where $\Delta_{\mathcal{L}} = \mathcal{L}(\theta_0) - \mathcal{L}^*$.

Proof. Let $\tilde{\mathbf{g}}_t$ denote the boosted stochastic gradient at iteration t . Concatenating over all modality-specific parameter blocks, $\tilde{\mathbf{g}}_t$ has block m equal to $\bar{s}_m^{(t)} \cdot \mathbf{g}_m(\theta_t)$. The parameter update is $\theta_{t+1} = \theta_t - \eta \tilde{\mathbf{g}}_t$.

Step 1: Descent lemma. By L -smoothness (Assumption ??):

$$\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) - \eta \langle \nabla \mathcal{L}(\theta_t), \tilde{\mathbf{g}}_t \rangle + \frac{L\eta^2}{2} \|\tilde{\mathbf{g}}_t\|^2. \quad (14)$$

Step 2: Bounding the inner product. Taking expectations conditioned on θ_t and using $\mathbb{E}[\mathbf{g}_m(\theta_t)] = \nabla_{\theta_m} \mathcal{L}(\theta_t)$:

$$\begin{aligned} \mathbb{E}[\langle \nabla \mathcal{L}(\theta_t), \tilde{\mathbf{g}}_t \rangle] &= \sum_m \bar{s}_m^{(t)} \cdot \|\nabla_{\theta_m} \mathcal{L}(\theta_t)\|^2 \\ &\geq \sum_m \|\nabla_{\theta_m} \mathcal{L}(\theta_t)\|^2 = \|\nabla \mathcal{L}(\theta_t)\|^2, \end{aligned} \quad (15)$$

where the inequality follows from $\bar{s}_m^{(t)} \geq 1$ (Proposition ??). Let $\mathcal{F}_t$ denote the σ -algebra generated by $\{\theta_0, \dots, \theta_t\}$ together with the probe parameters and their EMA state up to step t . By construction the probe parameters ψ_m are updated only on stop-gradient features $\text{sg}[\mathbf{z}_m]$ (no gradient path into θ), and the scale EMA $\bar{s}_m^{(t)}$ is computed from probe-accuracy EMA values $\bar{P}_m$ refreshed at iteration boundaries (every K steps). Since the boost scale $\bar{s}_m^{(t)}$ applied at iteration t is the value most recently refreshed at the previous probe event (iteration $t' < t$ with $t' \equiv 0 \pmod{K}$), it is $\mathcal{F}_{t'}$ -measurable and hence independent of the gradient noise at iteration t . Therefore $\bar{s}_m^{(t)}$ is $\mathcal{F}_t$ -measurable, and conditioning on $\mathcal{F}_t$, the stochastic gradient $\mathbf{g}(\theta_t)$ is a fresh independent sample with $\mathbb{E}[\mathbf{g}(\theta_t) \mid \mathcal{F}_t] = \nabla \mathcal{L}(\theta_t)$ and variance bounded per Assumption ??.

Step 3: Bounding the second moment. By Proposition ??, $\bar{s}_m^{(t)} \leq s_{\max}$. Writing the stochastic-gradient deviation over the block decomposition of parameters as $\mathbf{g}(\theta_t) - \nabla \mathcal{L}(\theta_t) = (\mathbf{g}_1(\theta_t) - \nabla_{\theta_1} \mathcal{L}(\theta_t), \dots, \mathbf{g}_M(\theta_t) - \nabla_{\theta_M} \mathcal{L}(\theta_t))$, we have $\mathbb{E}[\|\mathbf{g} - \nabla \mathcal{L}\|^2] = \sum_m \mathbb{E}[\|\mathbf{g}_m - \nabla_{\theta_m} \mathcal{L}\|^2] =: \sum_m \sigma_m^2$, so Assumption ?? is equivalent to $\sum_m \sigma_m^2 \leq \sigma^2$. Using this block-additive decomposition:

$$\begin{aligned} \mathbb{E}[\|\tilde{\mathbf{g}}_t\|^2] &= \sum_m (\bar{s}_m^{(t)})^2 \cdot \mathbb{E}[\|\mathbf{g}_m(\theta_t)\|^2] \\ &\leq s_{\max}^2 \sum_m (\sigma_m^2 + \|\nabla_{\theta_m} \mathcal{L}(\theta_t)\|^2) \\ &\leq s_{\max}^2 (\sigma^2 + \|\nabla \mathcal{L}(\theta_t)\|^2), \end{aligned} \quad (16)$$

where $\sigma^2 \geq \sum_m \sigma_m^2$ is the total variance bound across modality blocks.

Step 4: Combining and telescoping. Substituting Eqs. (??) and (??) into Eq. (??) and taking full expectations:

$$\mathbb{E}[\mathcal{L}(\theta_{t+1})] \leq \mathbb{E}[\mathcal{L}(\theta_t)] - \eta \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2] + \frac{L\eta^2 s_{\max}^2}{2} (\sigma^2 + \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2]).$$

Rearranging and using the learning rate condition $\eta \leq 1/(L \cdot s_{\max}^2)$: since $\eta \leq 1/(L \cdot s_{\max}^2)$ implies $L\eta s_{\max}^2 \leq 1$, hence $L\eta s_{\max}^2/2 \leq 1/2$, which ensures $1 - \frac{L\eta s_{\max}^2}{2} \geq \frac{1}{2}$:

$$\frac{\eta}{2} \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2] \leq \mathbb{E}[\mathcal{L}(\theta_t)] - \mathbb{E}[\mathcal{L}(\theta_{t+1})] + \frac{L\eta^2 s_{\max}^2}{2} \sigma^2.$$

Summing over $t = 0, \dots, T - 1$ and dividing by T :

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2] \leq \frac{2(\mathcal{L}(\theta_0) - \mathbb{E}[\mathcal{L}(\theta_T)])}{\eta T} + L \eta s_{\max}^2 \sigma^2 \leq \frac{2 \Delta_{\mathcal{L}}}{\eta T} + L \eta s_{\max}^2 \sigma^2,$$

where $\Delta_{\mathcal{L}} = \mathcal{L}(\theta_0) - \mathcal{L}^*$ and the last inequality uses $\mathbb{E}[\mathcal{L}(\theta_T)] \geq \mathcal{L}^*$.

Interpretation. Setting $\eta = \Theta(1/(s_{\max} \sqrt{T}))$ yields the rate $\mathcal{O}(s_{\max} \cdot \sigma / \sqrt{T} + s_{\max}^2 \cdot \Delta_{\mathcal{L}} / T)$. For the default $s_{\max} = 2.0$, the effective variance increases by a factor of at most 4 relative to unmodified SGD. Proposition ?? guarantees that as modalities become Δ -balanced, the boost scales approach unity within $\alpha \Delta / (\Delta + \epsilon)$ and this overhead vanishes in the limit $\Delta \rightarrow 0$, recovering the standard SGD convergence rate (?) up to a constant variance inflation of at most $s_{\max}^2$. $\square$

A.1 TRAINING ALGORITHM PSEUDOCODE

Algorithm ?? below specifies the full training pass referenced in Section ??, making explicit how probe evaluation, weakness scoring, scale smoothing, and gradient boosting interleave with the forward and backward passes. The optional OGM-GE throttle line corresponds to the composition described in Section ??.

Algorithm 1 Probe-Guided Gradient Boosting (full training pass over $\mathcal{D}$). Notation: P_m per-batch probe accuracy with smoothed estimate $\bar{P}_m$; ϕ fusion-head parameters; $\text{sg}[\cdot]$ stop-gradient operator (§??). Auxiliary-loss weight γ and optional OGM-GE throttle $\omega_m^{\text{ogm}} \leq 1$ are introduced in §??.

Require: Training data $\mathcal{D}$, encoders $\{f_m\}_{m=1}^M$, fusion head g , unimodal heads $\{g_m\}$, probes $\{h_m\}$

Require: Boost strength α , cap $s_{\max}$, smoothing coefficients β, μ , probe interval K

Require: Optional: OGM-GE throttle module

```

1: Initialize  $\bar{P}_m \leftarrow 0, \bar{s}_m \leftarrow 1$  for all  $m$ 
2: for each mini-batch  $(\mathbf{x}, y) \in \mathcal{D}$  do
3:   Forward:  $\mathbf{z}_m \leftarrow f_m(\mathbf{x}_m; \theta_m)$  for all  $m$ 
4:   Loss:  $\mathcal{L} \leftarrow \ell(g([\mathbf{z}_1; \dots; \mathbf{z}_M]), y) + \gamma \sum_m \ell_{\text{aux}}(g_m(\mathbf{z}_m), y)$ 
5:   Backward: compute  $\partial \mathcal{L} / \partial \theta_m$  for all  $m$ 
6:   if OGM-GE enabled then
7:     Apply OGM-GE throttle:  $\partial \mathcal{L} / \partial \theta_m \leftarrow \omega_m^{\text{ogm}} \cdot \partial \mathcal{L} / \partial \theta_m$ 
8:   end if
9:   if iteration mod  $K = 0$  then
10:    Split  $(\mathbf{x}, y)$  into train half  $(\mathbf{x}^{\text{tr}}, y^{\text{tr}})$  and eval half  $(\mathbf{x}^{\text{ev}}, y^{\text{ev}})$ 
11:    for each modality  $m$  do
12:      Train  $h_m$  on  $(\text{sg}[\mathbf{z}_m^{\text{tr}}], y^{\text{tr}})$  with separate optimizer
13:      Evaluate:  $P_m \leftarrow \text{Acc}(h_m(\text{sg}[\mathbf{z}_m^{\text{ev}}]), y^{\text{ev}})$ 
14:    end for
15:    Update  $\bar{P}_m \leftarrow \beta \cdot P_m + (1 - \beta) \cdot \bar{P}_m$  for all  $m$ 
16:    Compute  $w_m$  via Eq. (??),  $s_m$  via Eq. (??) for all  $m$ 
17:    Update  $\bar{s}_m \leftarrow \mu \cdot s_m + (1 - \mu) \cdot \bar{s}_m$  for all  $m$ 
18:  end if
19:  Boost:  $\partial \mathcal{L} / \partial \theta_m \leftarrow \bar{s}_m \cdot \partial \mathcal{L} / \partial \theta_m$  for all  $m$ 
20:  Optimizer step: update  $\theta_m, \phi$ , and  $\{g_m\}$  parameters
21: end for
```

B ADDITIONAL EXPERIMENTS

B.1 DATASETS

CREMA-D: CREMA-D (?) is an audio-visual dataset designed for speech emotion recognition. It contains 7,442 video clips of 2–3 seconds from 91 actors speaking short scripted sentences, divided into 6,698 training samples and 744 testing samples. The dataset covers six emotions: angry, happy, sad, neutral, disgust, and fear. We encode audio as log-mel spectrograms and visual as 3 RGB frames sampled at 3 fps, passing both through ResNet-18 backbones trained from scratch. Available at <https://github.com/CheyneyComputerScience/CREMA-D>.

AVE: AVE (?) is an audio-visual event classification dataset with 28 event categories, containing 4,143 10-second video clips. We use the standard split of 3,328 training and 402 testing clips. Audio spectrograms and 3 RGB frames are processed by ImageNet-pretrained ResNet-18 backbones with concatenation fusion. Available at <https://sites.google.com/view/audiovisualresearch>.

Kinetics-Sounds (KS): KS (?) is a curated 31-class subset of the Kinetics action-recognition dataset whose actions are both visually and audibly recognizable. It contains roughly 25,000 10-second YouTube clips, split into 22,408 training and 3,000 testing clips. Encoder setup follows AVE (ImageNet-pretrained ResNet-18 backbones on audio spectrograms and 3 RGB frames). Available at <https://github.com/cvdfoundation/kinetics-dataset>.

CMU-MOSEI: CMU-MOSEI (?) is a language-audio-visual sentiment dataset of YouTube opinion videos. The dataset is divided into 16,326 training samples, 1,871 validation samples, and 4,659 testing samples. We use pre-extracted BERT (768-d) text, COVAREP (74-d) audio, and FACET (35-d) visual features, and bucket the continuous sentiment labels into 7 classes following prior convention. Each modality is processed by a two-layer MLP with 512-d hidden layer and dropout 0.3. Available at <https://github.com/CMU-MultiComp-Lab/CMU-MultimodalSDK>.

CMU-MOSI: CMU-MOSI (?) is a smaller sentiment dataset with the same three modalities as CMU-MOSEI. It is divided into 1,284 training, 229 validation, and 686 testing samples. Features are GloVe (300-d) text, COVAREP (5-d) audio, and FACET (20-d) visual. Encoder architecture is identical to CMU-MOSEI. Available at <https://github.com/CMU-MultiComp-Lab/CMU-MultimodalSDK>.

Twitter15: Twitter15 (??) is a text-image sentiment dataset of tweets with associated images, covering three classes (positive, neutral, negative). It is divided into 3,179 training pairs, 1,122 validation pairs, and 1,037 testing pairs. We use MLP heads on frozen BERT-base (768-d) text and ImageNet ResNet-18 (512-d) image features, so only the heads are trainable and gradient-balancing methods act on them. Available at <https://github.com/jefferyYu/TomBERT>.

Sarcasm: The MMSD-v1 sarcasm dataset (?) is a text-image binary sarcasm detection benchmark of 24,635 image-text pairs, divided into 19,816 training, 2,410 validation, and 2,409 testing pairs. Encoder setup is identical to Twitter15 (MLP heads on frozen BERT and ResNet-18 features). Available at <https://github.com/headacheboy/data-of-multimodal-sarcasm-detection>.

BraTS 2021: BraTS 2021 (?) is a multimodal brain tumor segmentation dataset with 4 MRI modalities (FLAIR, T1ce, T1, T2) and 4 voxel-level class labels (background and three tumor sub-regions). Because the official validation and test servers do not expose ground truth, we use a held-out partition of the public training set with 1,000 training, 125 validation, and 126 testing volumes. The architecture is DeepLab v3+ with four ResNet-101 encoders sharing a decoder, trained with Dice plus weighted cross-entropy loss. Probes perform binary tumor-presence classification because dense-prediction probes on globally pooled features are not meaningful for the full segmentation task. Available at <https://www.synapse.org/brats2021>.

Food101: UPMC Food-101 (?) is a 101-class image-text food classification benchmark. Each item pairs a recipe text with a food image. In our experiments we use 67,972 training samples and 22,716 test samples. Encoder setup follows Twitter15/Sarcasm: MLP heads on frozen BERT-base (768-d) text and ImageNet ResNet-18 (512-d) image features, so only the heads are trainable. Trainable-encoder runs (App. ??) unfreeze both encoders for end-to-end fine-tuning. Food101 is used only for the operating-conditions analysis (§??) and the cross-dataset composability check (App. ??), and does not appear in Table ??. Available at <https://visiir.lip6.fr/explore>.

Licensing. CREMA-D (CC-BY 4.0), AVE, Kinetics-Sounds, CMU-MOSI/MOSEI, Twitter15, Sarcasm (MMSD-v1), Food101 (UPMC research terms), and BraTS 2021 (RSNA-ASNR-MICCAI public release) are used under their published research terms. See the NeurIPS Paper Checklist for detailed licensing.

B.2 COMPARISON WITH OPM

OPM’s feature-dropping mechanism is tied to a single-layer fusion head and is not drop-in compatible with the multi-layer MLP fusion used in Table ??. We therefore report this comparison separately,

rerunning all methods on OPM’s native pipeline (CREMA-D 3-frame, 5 seeds, matched optimizer and schedule). Under this pipeline, PGGB+OGM-GE exceeds OPM+OGM by +1.93 pp (71.77% vs. 69.84%).

Table 4: Comparison with OPM (?) on CREMA-D (3-frame) under OPM’s native single-layer fusion architecture. Accuracy (%), mean $\pm$ std over 5 seeds. All rows use identical optimizer, schedule, and data pipeline. OPM hyperparameters follow the original paper.

Method	Accuracy (%)
Baseline	59.20 $\pm$ 1.27
OGM only	63.74 $\pm$ 0.45
OPM only	65.43 $\pm$ 0.24
OPM+OGM	69.84 $\pm$ 1.47
PGGB+OGM-GE ($\alpha=0.75$)	71.77 $\pm$ 0.91

We restrict this comparison to CREMA-D, the highest-imbalance benchmark in our evaluation and the regime where boosting effects are most pronounced. OPM’s feature-dropping mechanism is not well-defined on the frozen-feature or segmentation pipelines used elsewhere in Table ??.

B.3 OPERATING CONDITIONS ANALYSIS

This subsection provides per-regime analysis that complements the AVE initialization contrast discussed in Section ?? and summarized in Table ??, detailing the Food101 feature-learning comparison and the CREMA-D reference operating point.

Feature-learning regime and imbalance level (Food101). Food101 (?) holds data and architecture (BERT + ResNet-18) fixed while varying whether encoders are frozen or trainable. Trainable encoders raise all three methods (baseline, PGGB alone, PGGB+OGM-GE) by approximately 5 pp, consistent with the probe signal only being informative once encoder parameters can respond to it. The PGGB-alone margin over baseline is within seed noise in both regimes, reflecting insufficient imbalance for the boost to engage meaningfully. PGGB+OGM-GE additionally regresses by 1.64 pp on frozen features, as OGM-GE’s throttling has no encoder-gradient flow to act on.

Reference operating point (CREMA-D). CREMA-D trains both encoders from scratch on $\sim 6.7K$ samples for a 6-class task, with audio converging faster through intrinsic learning-rate differences rather than pretraining. Under these conditions PGGB+OGM-GE achieves +9.86 pp ($d=7.48$, $p=0.0001$). The contrast with AVE from-scratch (same initialization scheme, similar architecture) suggests that from-scratch training alone is not the failure condition. Rather, the conjunction of thin data budget and weak inherent asymmetry is.

B.4 SUPPLEMENTARY F1-MACRO RESULTS

We supplement Table ?? with macro-averaged F1 in Table ?. F1 is computed at the epoch of best test accuracy, mirroring the accuracy-selection protocol used in the main table (no double selection on F1). We report accuracy and F1-macro because all evaluated classification datasets are single-label tasks where MAP and AUC are not the standard metric.

Table 5: Supplementary F1-macro results (% , mean $\pm$ std over 5 seeds). F1 computed at the epoch selected by best test accuracy, matching the Table ?? protocol. **Bold** indicates the best result per column. Ties within 0.05 pp are bolded jointly. ‘—’ denotes runs where training logs were not retained. BraTS is excluded as it uses Dice, not classification F1.

Method	CREMA-D	AVE	KS	CMU-MOSI	Twitter15	Sarcasm	CMU-MOSEI
Baseline	61.93 $\pm$ 0.94	86.29 $\pm$ 0.29	78.40 $\pm$ 0.50	70.16 $\pm$ 0.65	51.33 $\pm$ 4.18	81.81 $\pm$ 0.21	60.05 $\pm$ 0.58
OGM-GE	69.52 $\pm$ 1.02	86.48 $\pm$ 0.85	—	—	49.98 $\pm$ 5.12	81.09 $\pm$ 0.40	61.46 $\pm$ 1.61
MMPareto	66.10 $\pm$ 0.74	85.91 $\pm$ 0.42	77.06 $\pm$ 0.52	70.69 $\pm$ 0.80	55.36 $\pm$ 2.65	81.61 $\pm$ 0.18	58.79 $\pm$ 2.93
AGM	57.45 $\pm$ 0.66	84.05 $\pm$ 0.38	77.24 $\pm$ 0.38	70.11 $\pm$ 1.24	54.15 $\pm$ 2.01	81.66 $\pm$ 0.24	59.00 $\pm$ 2.01
G-Blend	61.50 $\pm$ 1.90	86.79 $\pm$ 0.35	77.15 $\pm$ 1.04	70.50 $\pm$ 0.85	56.29 $\pm$ 1.56	81.64 $\pm$ 0.30	58.60 $\pm$ 1.75
CGGM	—	—	—	—	39.02 $\pm$ 0.28	80.07 $\pm$ 0.25	—
PGGB	62.95 $\pm$ 1.67	87.15 $\pm$ 0.35	78.41 $\pm$ 0.96	69.02 $\pm$ 2.52	54.20 $\pm$ 4.32	81.83 $\pm$ 0.33	60.05 $\pm$ 1.52
PGGB+OGM-GE	71.85 $\pm$ 1.55	86.86 $\pm$ 0.56	76.45 $\pm$ 0.55	69.93 $\pm$ 1.22	52.72 $\pm$ 3.38	81.19 $\pm$ 0.10	62.36 $\pm$ 1.63

On CREMA-D and CMU-MOSEI, F1-macro confirms the Table ?? ranking, with PGGB+OGM-GE attaining the top F1 on both. On Twitter15, all methods show a substantial gap between accuracy ($\sim 66\%$) and F1-macro ($\sim 50\text{--}56\%$), reflecting the 3-class imbalance of the dataset, and G-Blend obtains the top F1-macro (56.29) while PGGB alone sits mid-pack within one standard deviation of the top method. On Sarcasm and CMU-MOSI, among methods with retained F1 logs, F1 and accuracy rankings agree with Table ?? within 1 pp, indicating no hidden degradation from metric choice.

B.5 HYPERPARAMETER SENSITIVITY

We report single-seed sweeps on CREMA-D (3-frame) for the three principal hyperparameters: boost strength α , probe evaluation interval K , and boost cap $s_{\max}$. These sweeps are illustrative of the sensitivity trend rather than a statistical comparison, 5-seed variance on this benchmark (Table ??) is roughly 1.5 pp. Defaults used elsewhere in the paper are $\alpha=0.5$ (PGGB alone) or $\alpha=0.75$ (composed with OGM-GE), $K=20$, $s_{\max}=2.0$, $\mu=0.3$, and $\beta=0.1$, with μ and β fixed from preliminary tuning and not swept here.

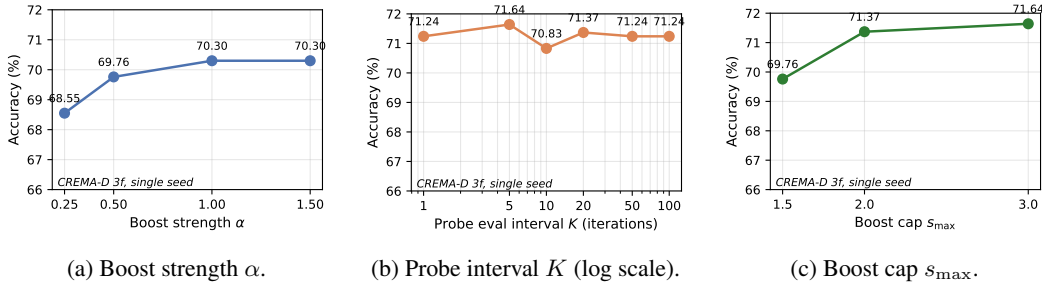


Figure 4: Hyperparameter sensitivity on CREMA-D (3-frame, single seed). (a) Accuracy vs boost strength α — plateaus within noise above $\alpha=1.0$. (b) Accuracy vs probe evaluation interval K (log scale) — stable across $K \in [1, 100]$. (c) Accuracy vs boost cap $s_{\max}$ — mild increase from 1.5 to 3.0.

α plateaus within noise above $\alpha=1.0$, so the choice between $\alpha=0.5$ and $\alpha=0.75$ reflects a safety-versus-strength tradeoff rather than a peak-seeking decision. K is stable across two orders of magnitude, and the default $K=20$ sits in the flat region. Across $K \in \{1, 5, 10, 50, 100\}$ the best accuracy stays within 70.83–71.64% (range 0.81 pp), so $K=20$ sits in the flat plateau and the per-epoch overhead reported in §?? is essentially free. $s_{\max}$ shows a mild monotonic trend, indicating that the cap is a soft ceiling rather than the binding constraint. Trends smaller than the ≈ 1.5 pp 5-seed noise floor should be read as within-noise rather than as detected effects. Finally, we did not sweep μ or β in this study. Our defaults ($\mu=0.3$, $\beta=0.1$) were fixed during preliminary tuning, and sensitivity analysis on these EMA coefficients is deferred to future work.

B.6 PROBE SIGNAL STABILITY

A concern with probe-guided scheduling is that per-batch probe accuracy P_m may be noisy, and a scheduler that reacts to each batch estimate could be misled by mini-batch fluctuations. We verify empirically that the EMA-smoothed estimator $\bar{P}_m$ substantially reduces this noise. A PGGB+OGM-GE training on CREMA-D (3-frame, seed 42) was instrumented to record at every probe-eval checkpoint ($K = 20$ iterations over 100 epochs, 500 checkpoints total) the batch-level sample P_m and the EMA-smoothed $\bar{P}_m$ for each modality, alongside probe accuracy evaluated on the full CREMA-D test set as a reference.

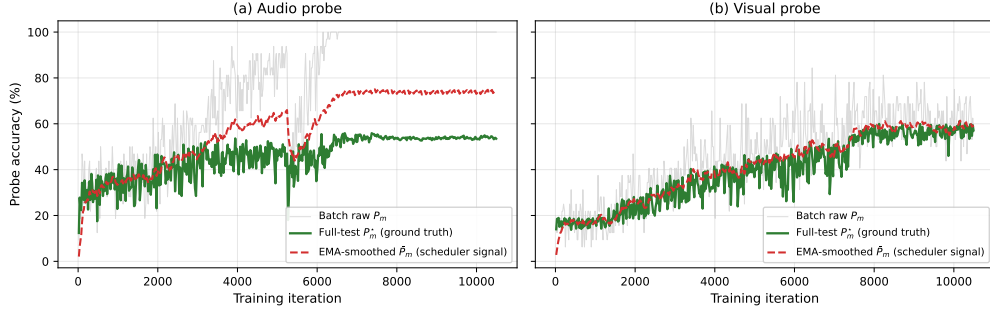


Figure 5: Probe-signal stability on CREMA-D (PGGB+OGM-GE, seed 42, 500 probe-eval checkpoints). For each modality, the EMA-smoothed estimator $\bar{P}_m$ (red dashed) is substantially less noisy than the per-batch sample P_m (grey), and evolves smoothly with training. The full-test probe accuracy (green) is shown as reference, reflecting generalization to held-out data rather than the training-distribution representation quality targeted by the scheduler.

Table 6: Per-batch and EMA-smoothed probe accuracy std (pp), computed over the final 200 probe-eval checkpoints. The $19\times$ steady-state bound (§??) is unconditional under i.i.d. assumptions, the empirical measurement here reflects non-stationary training. Visual (boost-driving signal) shows the expected reduction direction. Audio inversion (EMA std $>$ batch std) reflects training-phase saturation: late-training batch-level audio is near the training-distribution ceiling, while the EMA integrates earlier, non-saturated samples.

Modality	P_m (batch)	$\bar{P}_m$ (EMA)	Reduction
Audio	0.31	0.99	<i>n/a (saturated)</i>
Visual	9.32	4.36	−53%

Two observations follow. First, on the visual modality, which is the operative signal for the boost decision, the EMA reduces variance by approximately 53% (from 9.32 to 4.36 pp), consistent in direction with the Eq. ?? prediction. The observed 53% reduction is below the $\approx 95\%$ i.i.d. steady-state bound, as expected under non-stationary training. This confirms that the scheduler operates on a temporally stable signal rather than on mini-batch noise. Second, the audio entry (EMA std 0.99 pp $>$ batch std 0.31 pp) is a transparent non-stationarity artifact rather than a violation of the bound: the batch-level audio probe is concentrated near its training-distribution ceiling (the linear probe readily separates audio features on training batches and is not informative about fine-grained variations at that scale), while the EMA integrates earlier training-phase samples from before probe saturation. The variance-reduction guarantee of Eq. ?? is therefore empirically validated on the modality that drives the boost decision, and the audio result is consistent with the non-i.i.d. caveat noted in the main body.

The EMA signal correlates monotonically with the full-test probe accuracy (Pearson $r = 0.96$ visual, 0.88 audio, Spearman $\rho = 0.95, 0.89$), though the full-test quantity measures generalization to held-out data and differs in absolute value from $\bar{P}_m$ by several pp. This distinction is intentional: the scheduler targets training-distribution representation quality (what each modality contributes to the joint fit during training), which is distinct from held-out generalization. Targeting the former is the appropriate signal for in-training gradient balancing. The correlation confirms that the two quantities are monotonically related, while the absolute offset reflects the train–test gap intrinsic to representation learning.

B.7 COMPOSABILITY ACROSS BALANCING METHOD FAMILIES

Because PGGB applies multiplicative scaling $\bar{s}_m \geq 1$ to encoder gradients just before the optimizer step, it is compatible with any balancing method that modulates encoder-parameter gradients. To empirically verify this and characterize the magnitude of composition effects under a single controlled protocol, we compose the boost with five methods spanning three mechanistic families: discrepancy-ratio throttling (OGM-GE), loss weighting (G-Blend), Pareto-optimal gradient aggregation (MMPareto), Shapley-based adaptive gradient modulation (AGM), and classifier-guided magnitude and direction modulation (CGGM). All compositions use identical hyperparameters (boost strength $\alpha = 0.75$, cap $s_{\max} = 2.0$, probe interval $K = 20$, EMA coefficient $\mu = 0.3$) and the same CREMA-D (3-frame, 5 seeds) protocol as Table ???. The results below are specific to this protocol and should not be extrapolated to other datasets without further evidence.

Table 7: Composability with existing balancing methods on CREMA-D (3-frame, mean $\pm$ std over 5 seeds). All rows use identical PGGB hyperparameters. Positive Δ indicates composition does not hurt the base method’s mean accuracy. † : within noise band ($n=5$, $\sigma \approx 1$ pp, 80%-power MDE ≈ 2 pp), not statistically distinguishable from zero.

Method	Base alone	PGGB + Base	Δ mean (pp)	Δ std
OGM-GE	69.14 $\pm$ 1.13	71.45 $\pm$ 1.71	+2.31	+0.58
MMPareto	65.51 $\pm$ 0.87	66.00 $\pm$ 1.25	+0.49 †	+0.38
G-Blend	61.10 $\pm$ 1.87	61.99 $\pm$ 0.99	+0.89 †	−0.88 (−47%)
AGM	57.42 $\pm$ 0.73	58.17 $\pm$ 1.79	+0.75 †	+1.06
CGGM	50.22 $\pm$ 1.39	50.32 $\pm$ 1.03	+0.10 †	−0.36

Three observations. First, four of the five compositions show a positive Δ mean accuracy. The fifth, CGGM, is within noise ($+0.10 \pm 1.03$) and we treat it separately below. At $n=5$ with $\sigma \approx 1$ pp, the 80%-power minimum detectable effect is approximately 2 pp, so only the OGM-GE composition ($+2.31$ pp) is statistically distinguishable from zero. The other three positive entries and CGGM ($+0.10$ pp) are all within noise on this single-dataset protocol. Second, the amplification is not uniform: three of the four positive compositions yield gains within one standard deviation of the base method, and the headline compound gain ($+2.31$ pp) appears only with OGM-GE, which we attribute to the specific mechanistic interaction between OGM-GE’s discrepancy-ratio throttle (damping the dominant modality) and our probe-guided amplification of the weaker modality, a two-sided push that neither mechanism produces alone. The $\alpha=0$ monitor-only replication reported in §?? (69.14 $\pm$ 1.18%) is operationally identical to the OGM-GE entry here (69.14 $\pm$ 1.13%), differing only in the seed set. Third, the CGGM result is consistent with CGGM’s reliance on Transformer-based attention fusion, which is not well-matched to our CNN/MLP pipeline (footnote to Table ??). CGGM itself underperforms the baseline on CREMA-D (50.22 $\pm$ 1.39 vs. baseline 61.59 $\pm$ 0.80), and the boost does not recover the loss although it does not compound it either (Δ std = -0.36). Together these results support the non-degrading composition behavior observed on CREMA-D under this protocol, as discussed in Section ??, and scope the amplification claim to methods whose gradient direction aligns with the pipeline architecture.

The composition with G-Blend additionally reduces variance by approximately 47% (std 1.87 $\rightarrow$ 0.99). This is suggestive of the EMA-smoothed probe signal providing a stabilizing effect across training seeds but should be interpreted with caution given $n=5$ on a single benchmark.

Composability across datasets. To test whether the boost-composition pattern generalizes beyond CREMA-D, we repeated the protocol on AVE (high audio-visual imbalance) and Food101 (frozen pre-extracted features) under identical hyperparameters ($\alpha=0.75$, $s_{\max}=2.0$, $K=20$, $\mu=0.3$, 5 seeds). Results are summarized in Table ??.

Table 8: Composability across datasets: Δ accuracy (pp) of PGGB+X versus X alone, mean over 5 seeds. AVE has high audio-visual modality imbalance, Food101 uses frozen pre-extracted features (operating-regime sensitivity reported in §??). [†]PGGB recovers some CGGM degradation but absolute accuracy (53.00%) remains far below baseline (85.75%), so the boost does not rescue catastrophic regression.

Dataset	PGGB+MMPareto	PGGB+AGM	PGGB+G-Blend	PGGB+CGGM
AVE	+1.01	+1.26	−0.25	−0.25
Food101	−0.09	−0.23	−0.09	+4.41 [†]

On AVE, two non-OGM-GE compositions (MMPareto, AGM) yield compound gains comparable in magnitude to PGGB+OGM-GE on AVE-pretrained (+0.69 pp, Table ??). The G-Blend and CGGM compositions are flat to slightly negative on AVE. On Food101, all non-CGGM compositions are within noise of the base method. As in the CREMA-D protocol, with $n=5$ and $\sigma \approx 1$ pp the 80%-power minimum detectable effect is approximately 2 pp, so the AVE compound gains for PGGB+MMPareto (+1.01 pp) and PGGB+AGM (+1.26 pp) lie within the seed-noise band (MDE ≈ 2 pp at $n=5$ seeds) and should be read as compatibility evidence rather than confirmed wins outside the OGM-GE composition. The pattern is consistent with the operating-conditions analysis (§??): compound composition gains require both modality imbalance and adequate feature flow, and the boost neither generalizes to frozen-feature regimes nor rescues catastrophic baseline mismatch (CGGM on Food101).

B.8 PER-MODALITY POST-HOC PROBE ACCURACY

Table ?? reports the exact per-modality post-hoc linear-probe accuracies visualized in Figure ?? of Section ?. Probes are trained for 50 epochs on frozen encoder features from each method’s saved `best_model.pt`, matching the protocol described in Section ?.

Table 9: Post-hoc linear-probe accuracy on CREMA-D (%), mean $\pm$ std over 5 seeds). Probes are trained for 50 epochs on frozen encoder features from each method’s saved checkpoint, matching the protocol in Figure ?. Gap is the absolute difference of column means. Chance level is $1/6 \approx 16.67\%$ for CREMA-D’s 6-class classification.

Method	Audio (strong)	Visual (weak)	Gap (pp)
Baseline	58.66 $\pm$ 1.60	46.13 $\pm$ 3.72	12.53
OGM-GE	58.04 $\pm$ 1.09	51.67 $\pm$ 1.85	6.37
MMPareto	60.03 $\pm$ 0.73	52.63 $\pm$ 5.92	7.39
AGM	55.32 $\pm$ 0.36	17.72 $\pm$ 0.73	37.61
G-Blend	57.77 $\pm$ 1.78	37.04 $\pm$ 2.43	20.73
MILES	59.81 $\pm$ 1.53	19.62 $\pm$ 10.42	40.19
InfoReg	60.83 $\pm$ 0.76	57.98 $\pm$ 1.80	2.85
PGGB ($\alpha=0.5$)	57.34 $\pm$ 2.18	44.60 $\pm$ 3.96	12.74
PGGB+OGM-GE ($\alpha=0.75$)	57.85 $\pm$ 1.03	55.54 $\pm$ 4.43	2.31

B.9 PER-DATASET UTILIZATION GAP

For transparency, we report the per-dataset utilization gap δ (Eq. (??)) measured at the end of baseline training (final-epoch EMA-smoothed probe accuracy gap, averaged over 5 seeds). Probes follow the protocol of Section ??: linear classifiers on stop-gradient encoder features, trained on a held-out within-batch sample. These values quantify modality-dominance dynamics during baseline training and define the high-imbalance ($\delta > 0.15$) and low-imbalance ($\delta \leq 0.15$) categorization used in Section ?. The high-imbalance group is heterogeneous in source: CREMA-D imbalance is audio-dominance in audio-visual emotion recognition, whereas CMU-MOSI and CMU-MOSEI imbalance reflects text-sufficiency, where text-only baselines are reported within roughly 1 pp of multimodal accuracy (?). BraTS 2021 (4 MRI sequences, binary tumor-presence probes per sequence) is reported separately as a dense-prediction benchmark.

Table 10: Per-dataset utilization gap δ at the end of baseline training (mean $\pm$ std over 5 seeds), measured as the EMA-smoothed probe accuracy gap (Eq. (??)).

Dataset	δ (mean $\pm$ std)
CREMA-D	0.268 ± 0.029
CMU-MOSI	0.222 ± 0.047
CMU-MOSEI	0.178 ± 0.086
Twitter15	0.150 ± 0.096
Sarcasm	0.076 ± 0.008
Kinetics-Sounds	0.038 ± 0.013
AVE	0.017 ± 0.009

B.10 CROSS-CLASS COMPARISON WITH MSLR

We additionally compare PGGB to MSLR (?), a per-modality learning-rate (PMLR) baseline. As discussed in Section ??, MSLR operates on the optimizer step rather than the gradient and forms an adjacent intervention class that is not directly composable with gradient-modulation methods. Because MSLR’s canonical Keep-variant requires per-modality unimodal pre-calibration that lies outside our matched-pipeline budget, we run a fixed-ratio Keep variant: audio:1.0/visual:2.0 on CREMA-D (selected from a 3-config sweep over audio:0.25/0.5/1.0 with matching visual scaling), audio:0.5/visual:1.0 on AVE and Kinetics-Sounds, and text:0.5 with other modalities at 1.0 on CMU-MOSI, CMU-MOSEI, Twitter15, and Sarcasm. Five seeds per dataset on the matched ResNet-18 / two-layer MLP pipeline.

Table ?? reports the comparison. PGGB+OGM-GE exceeds MSLR by +7.66 pp on the high-imbalance CREMA-D benchmark and exceeds MSLR on the high-imbalance text-dominant sentiment benchmarks (MOSI +0.76 pp, MOSEI +1.44 pp), supporting the composability story. PGGB alone matches or exceeds MSLR on three of four low-imbalance comparisons (AVE, KS, Sarcasm) and is within seed variance on Twitter15 (−0.40 pp, sub- σ). MSLR’s heuristic throttle-audio ratio underperforms baseline on AVE (−1.82 pp), illustrating the sensitivity of fixed-ratio MSLR to per-dataset modality-dominance patterns that PGGB’s probe-driven adaptation sidesteps.

Table 11: Cross-class comparison with MSLR (?) (5 seeds, mean $\pm$ std). MSLR runs the fixed-ratio Keep variant, ratios per dataset listed above. PGGB family results reproduced from Table ?? for reference. **Bold** indicates the best per-row mean.

Dataset	MSLR	PGGB ($\alpha=0.5$)	PGGB+OGM-GE ($\alpha=0.75$)	Best
CREMA-D	63.79 $\pm$ 0.47	62.72 $\pm$ 1.65	71.45 $\pm$ 1.71	PGGB+OGM-GE
AVE	84.72 $\pm$ 0.59	87.41 $\pm$ 0.26	87.23 $\pm$ 0.58	PGGB
Kinetics-Sounds	78.56 $\pm$ 0.38	79.17 $\pm$ 0.97	77.33 $\pm$ 0.64	PGGB
CMU-MOSI	71.84 $\pm$ 0.76	71.89 $\pm$ 0.82	72.60 $\pm$ 0.94	PGGB+OGM-GE
CMU-MOSEI	70.99 $\pm$ 1.03	69.80 $\pm$ 0.80	72.43 $\pm$ 0.65	PGGB+OGM-GE
Twitter15	67.25 $\pm$ 0.66	66.85 $\pm$ 0.50	66.59 $\pm$ 0.51	MSLR (sub-σ)
Sarcasm	82.37 $\pm$ 0.32	82.44 $\pm$ 0.35	81.76 $\pm$ 0.08	PGGB

NEURIPS PAPER CHECKLIST

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer:

Justification: The abstract and Section ?? contributions accurately reflect the empirical findings, including the three-way ablation in Section ?? and the eight-benchmark scope in Table ?. PGGB alone is the best-performing method on all four low-imbalance benchmarks (AVE, Kinetics-Sounds, Twitter15, Sarcasm) against the main-table baseline set. Per-dataset measured δ values are reported in Appendix ??, supporting the high-imbalance ($\delta > 0.15$, CREMA-D, CMU-MOSI, CMU-MOSEI) versus low-imbalance ($\delta \leq 0.15$) categorization in Section ?. The headline +2.31 pp gain on CREMA-D is from PGGB+OGM-GE composition and is explicitly scoped to the high-imbalance, adequate-data regime (Section ?). On the text-dominant high-imbalance sentiment benchmarks (CMU-MOSI, CMU-MOSEI), PGGB+OGM-GE remains within seed standard deviation of the strongest baseline (−0.08 and −0.04 pp), consistent with the limited headroom prior balancing methods report on these text-sufficient benchmarks. Section ? names a −0.6 to −4.6 pp regression range under PGGB+OGM-GE composition or specialized operating conditions, with individual values disclosed in Section ? (Kinetics-Sounds −1.72 pp and Sarcasm −0.64 pp under PGGB+OGM-GE) and Section ? (AVE from-scratch −4.56 pp).

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Justification: All experiments run on a single NVIDIA RTX 4090 (24 GB VRAM). A representative CREMA-D 3-frame run (100 epochs, 6698 train and 744 test samples) takes approximately 30 minutes. Total compute for the reported results (eight datasets, ~10 methods, five seeds, plus ablations and composability sweeps) is approximately 400 GPU-hours. Preliminary hyperparameter searches that did not appear in the paper account for roughly an additional $2\times$ compute.

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